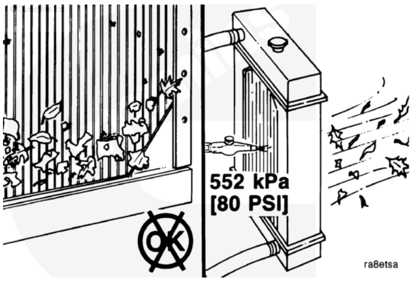
Trainings

Initial Check

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



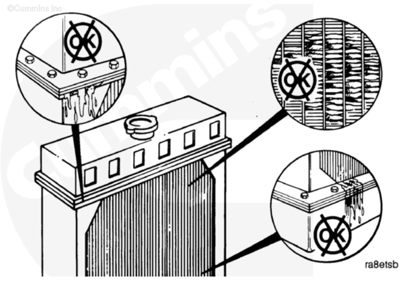
Check for plugged radiator fins.

Use compressed air to blow out the dirt and debris. Blow the compressed air in the opposite direction of the fan air flow.

Measurements		
	kpa	psi
Air Pressure	552	80

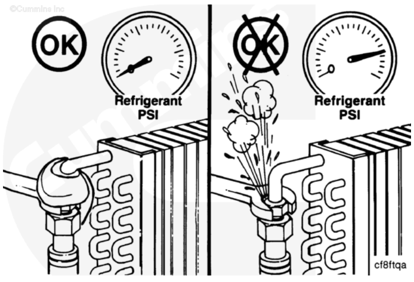
Check for bent or broken radiator fins. Check for radiator core and gasket leaks.

If the radiator **must** be replaced, refer to the manufacturer's replacement procedures.



⚠ WARNING ⚠

If a liquid refrigerant system (air conditioning) is used, wear eye and face protection, and wrap a cloth around the fitting before removing. Liquid refrigerant can cause serious eye and skin injuries.



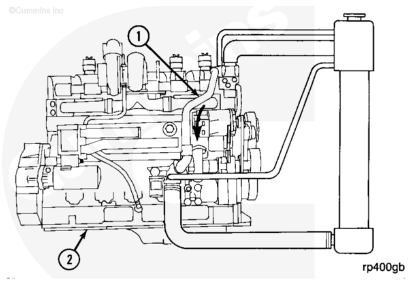
⚠ WARNING ⚠

To protect the environment, liquid refrigerant systems must be properly emptied and filled using equipment that prevents the release of refrigerant gas into the atmosphere. Federal

law requires capturing and recycling the refrigerant.

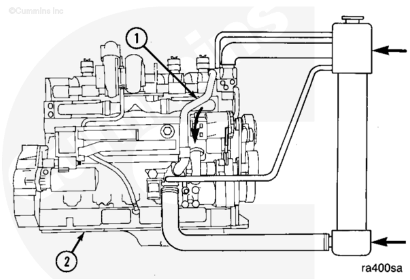
For environmental protection, federal regulations require that Freon be recycled, and **not** vented into the atmosphere.

Reverse Flow Test

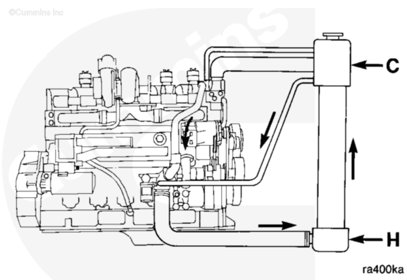


This test **must** be performed with the cooling system completely full and the coolant at ambient temperature.

Operate the engine at medium to high idle. Since the thermostats are closed, coolant flow is through the bypass line (1) and the engine (2).



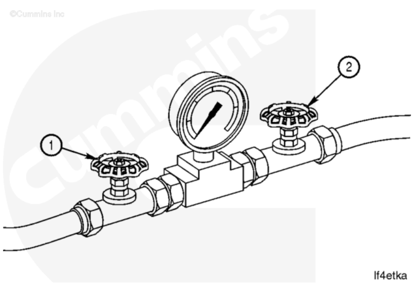
As the engine is warming up, use a contact pyrometer to check the temperature of the radiator top and bottom tanks.



If the bottom tank begins to warm and the top tank remains cold, the radiator has reverse flow. Refer to manufacturer's instructions for repair procedures.

Check the piping to make sure it is the correct size and routed to specifications. Refer to the OEM service manual for installation instructions.

Test

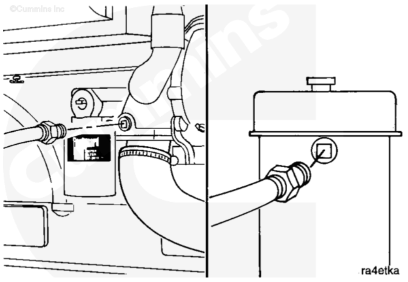


Use a pressure gauge designed to measure 0 to 100 kPa [0 to 15 psi] to measure obstruction.

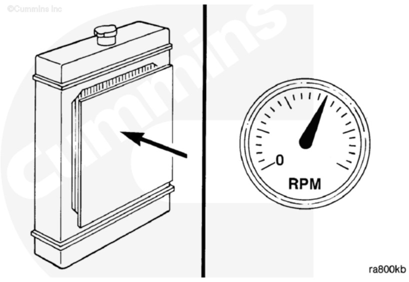
Install a shutoff valve (1)(2) on each side of the gauge.

Install one line into the water pump inlet connection and the other end into the thermostat return line on top of the radiator core.

Vent the air from the gauge lines.



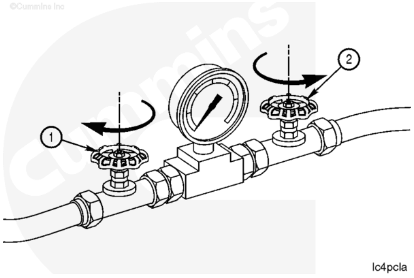
Restrict the radiator air flow. Operate the engine until the coolant temperature reaches 88 to 90°C [190 to 194°F].



Operate the engine at rated rpm.

Close valve (1) and open valve (2).

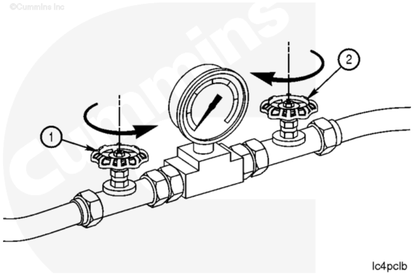
Record the pressure at the top of the radiator.



Close valve (2) and open valve (1).

Record the pressure at the water pump inlet.

Compare the two pressure readings. If they differ by more than 35 kPa [5 psi], the radiator core or piping is obstructed.



Last Modified: 15-May-2003